Modern Application Development I Project Report

1. Student Details

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About Me: I am a diploma student at IIT Madras BS Degree program passionate about web application development and database management. I enjoy implementing real-world projects that strengthen my technical knowledge and practical development skills.

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2. Project Details

Project Title: PeakTrail - A Trekking Management Application

Problem Statement:

To build a trekking management web application that allows **Admin, Trek Staffs, and Users (Trekkers)** to interact with the system based on their roles, thus simplifying the management of trekking expeditions.

Approach:

The application was developed using Flask as the backend framework with a modular structure. The system implements role-based authentication to provide separate dashboards and functionalities for Admin, Trek Staffs, and Trekkers. It allows trek scheduling, staff assignment, online bookings and participant management through a centralized platform.

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3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in debugging, improving frontend responsiveness and preparing parts of the project documentation.
The extent of AI/LLM usage is around **10–12%**, limited to **code suggestions and documentation**.
All project design decisions, final implementation logic, testing, and integration were completed manually.

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4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web application framework
SQLAlchemy	Object Relational Mapper (ORM) for SQLite database
SQLite	Database for storing application data
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
HTML5	Structure of web pages
CSS3	Styling and responsive layouts
Flask-Login	User authentication and session management

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5. Database Schema / ER Diagram

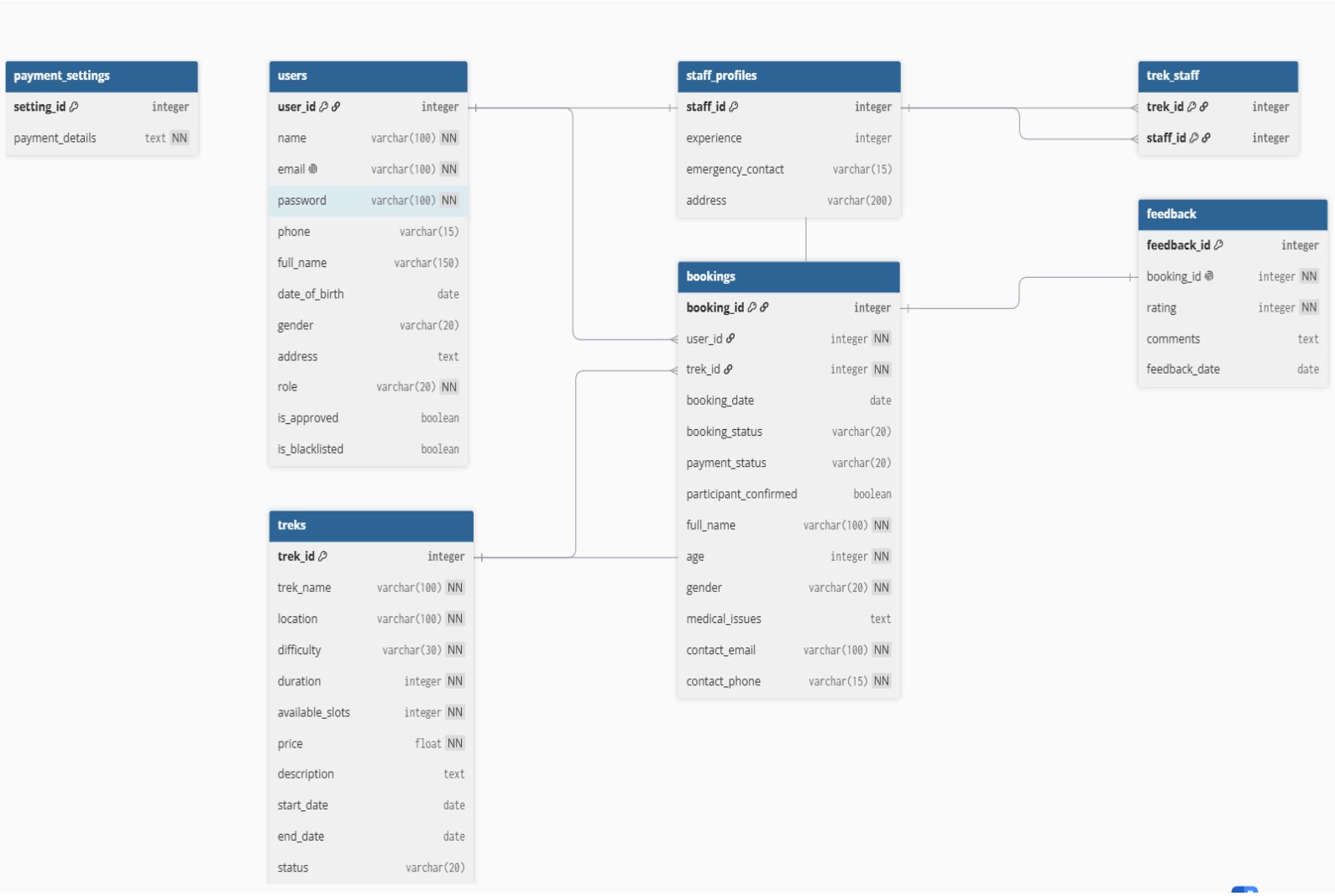
Tables:

1. **Users** — Stores the personal information, login credentials, and account status of all users (Trekkers, Trek Staff, and Admin). (user_id, name, email, password, phone, full_name, date_of_birth, gender, address, role, is_approved, is_blacklisted)
2. **Staff_Profile** — Stores additional details specific to Trek Staff members (staff_id, experience, emergency_contact, address)
3. **Trek** — Stores information about all trekking expeditions offered by the organization (trek_id, trek_name, location, difficulty, duration, available_slots, price, description, start_date, end_date, status)
4. **Trek_Staff** — Association table that maps Trek Staff members to the treks they are assigned to, implementing a many-to-many relationship (trek_id, staff_id)
5. **Bookings** — Stores booking records made by users for treks along with participant and payment details (booking_id, user_id, trek_id, booking_date, booking_status, payment_status, participant_confirmed, full_name, age, gender, medical_issues, contact_email, contact_phone)
6. **Feedback** — Stores ratings and reviews submitted by users for completed trek bookings (feedback_id, booking_id, rating, comments, feedback_date)
7. **Payment_Settings** — Stores the payment instructions or payment details configured by the administrator (setting_id, payment_details)

Relationships:

- One-to-One → **Users** → **Staff Profiles**
- One-to-Many → **Users** → **Bookings**
- One-to-Many → **Treks** → **Bookings**
- One-to-One → **Bookings** → **Feedback**
- Many-to-Many → **Staff Profiles** ↔ **Treks** (via Trek_Staff)

ER diagram (created using dbdiagram.io) :



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6. Architecture and Features

Architecture Overview:

- **app.py** – Main Flask application entry point
- **models.py** – SQLAlchemy database models

- **routes.py** – Route definitions and application logic
- **extensions.py** – Initializes Flask extensions such as SQLAlchemy and Flask-Login
- **templates/** – Jinja2 HTML templates for all user interfaces
- **static/** – CSS and image resources

Implemented Features:

- Role-based authentication for Admin, Trek Staff, and Trekkers.
- User registration and secure login.
- Admin management of trekking expeditions and tracking booking records.
- Trek Staff assignment to expeditions.
- Online trek booking with participant details.
- Booking approval and payment status management.
- Trek status and participant management.
- Responsive user interface for desktop, tablet, and mobile devices.

Additional Features:

- Configurable payment instructions managed by the Admin.
- Feedback submission and ratings by trekker after trek completion.

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7. Video Presentation

Drive Link:
https://drive.google.com/file/d/1CL9QkaDZLqC_zhcVrMRQBGx5bAEGv8Iq/view?usp=sharing